

Hylton Reber

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hyltonreber.github.io

OBJECTIVE

Mechanical engineering graduate seeking an entry-level role to apply design, analysis, and problem-solving skills to support impactful engineering projects and deliver high-quality results within a collaborative team.

PRACTICAL EXPERIENCE

Hyster-Yale Forklift Overhead Guard | Capstone Senior Team Project Aug 2024 – May 2025

- Reduced resonance noise in an overhead guard to contribute to R&D and improve the user experience.
- Integrated accelerometers via Arduino to perform modal testing.
- Interpreted accelerometer sensor specification sheets and configured 8-bit registers in C++.
- Developed a physics-based testing method to determine the natural frequency for modal analysis.

Student Manager | University of Idaho Outdoor Program – Climbing Center Jan 2023 – May 2025

- Wrote and edited dozens of pages of operational plans and provided ongoing support and development for staff.
- Provided excellent customer service, safety enforcement, rule adherence, and facility maintenance to ensure a secure and welcoming environment for patrons and employees.
- Facilitated over 50 clinics to educate patrons on safety measures and advanced climbing skills.

Refrigeration Cycle Team Project | Senior Laboratory Aug 2024 – Dec 2024

- Conducted experimental analysis on a refrigeration system, collecting 20+ measurement cycles of temperature, pressure, and airflow across condenser and evaporator.
- Applied thermodynamic principles and state postulate analysis with EES and MATLAB to determine system heat transfer and efficiency.

Axes and Axles LLC | Team Lead | Red Bull Soapbox Race Denver 2026 Apr 2026 – Jun 2026

- Led 7-person team rebuilding chassis to meet competition weight, clearance, and performance requirements.
- Redesigned front suspension twice to balance ground clearance against weight limits, relocating the steering arm to correct a steering inversion introduced by the rework.

SolidWorks Steam Engine Full Assembly | SolidWorks Course Jan 2025 - May 2025

- Modeled and assembled a full steam engine in SolidWorks, producing detailed part drawings, renderings, and assembly animations while applying design documentation, tolerancing, and configuration control per engineering drawing standards.

Machine Shop Experience | UI Machine Shop Aug 2023 – May 2025

- Accumulated 100+ hours of hands-on machining on manual and CNC mills and lathes.
- Fabricated components (nested cube, tap holder) demonstrating proficiency in measurement, tolerancing, and multi-axis setups.

EDUCATION: UNIVERSITY OF IDAHO, MOSCOW, ID

Bachelor of Science | Mechanical Engineering

May 2025

GPA: 3.63

SKILLS AND CERTIFICATIONS

Certifications: FE Exam – Passed; Certified SolidWorks Expert (CSWE); Aerospace Academic Certificate.

Proficient: Microsoft Office, MATLAB, Python, EES, CAD, Technical Drawings, and Technical Writing.

Technical Electives: Fatigue and Fracture Mechanics, Gas Dynamics, Turbomachinery, Aerodynamics.

CLUBS AND ORGANIZATIONS

Risk Manager | Phi Kappa Tau Aug 2022 – May 2025

Aeromechanical Member | Vandal Atmospheric Science Team (VAST) Sep 2021 – May 2024

Eagle Scout | Scout Troop 310 – Idaho Falls, ID 2013 – 2020